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Data Science FINAL PROJECT REPORT

Project Title

COVID-19 DETECTION USING CHEST X-RAY IMAGES USING CNN

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GITHUB LINK :
https://github.com/RAVIAPTISWETHA/COVID-19-Detection-from-Chest-X-ray-Images-Using-CNN-.git

DECLARATION STATEMENT

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COVID-19 Detection from Chest X-ray ImagesUsing CNN

Abstract

The survey looks at allocation of chest X ray images into three groups using deep learning(DL) addresses : COVID19 Normal & Pneumonia In addition to enhancing model interpretability with Grad-Cam the study attempts to assess the effiency of convolutional neural networks(CNNs) medical image allocation.

This dataset was preprocessed by removing duplicate images verifying data quality and applying data augmentation technique to improve model generalisation a baseline CNN model was developed and compared with a tuned CNN and a transfer learning model based on Dense Net121 fine tuning was further tested to build up the achievement of the transfer learning model class weights were also used to address class imbalance during traning.

The results shows that the Baseline CNN accomplished the biggest test accuracy of 97.47% surpass both tuned CNN (89.08%) and DenseNet121 model(96.01%)fine tuning improved the DenseNet121 model to 96.54%although it did not exceed the performance of the Baseline CNN these findings suggest that simpler models can perform effectively when supported by appropriate preprocessing and training strategies.

Grad-Cam visualisation identified that the models into clinically relevant place of the lungs improving the interpretability of predictions Overall the class feature usefulness of data quality model collection and explainability in developing reliable deep learning (DL) systems for medical image analysis

Chapter 1 : Introduction

1.1 Introduction

Medical imaging plays a major role in latest health care particularly in diagnosis & monitoring of respiratory diseases among the available imaging techniques chest X ray are widely used due to their accessibility low cost and effectiveness in identifying conditions Pneumonia & Covid 19 however interpreting chest X ray images require significant clinical expertise and the increasing demand for rapid diagnosis can place a considerable burden on radiologists as long as viable interest in developing automated systems that can support medical professionals in analysing these images more efficiently and accurately

In recent years artificial intelligence (AI) & more individually DL has demonstrated strong potential in the ground of medical images analysis CNNs have converted the demonstrative coming image classification task payable to their strength to acquire stratified quality from raw image information (Lecun et al.,1998) the success of deep CNN architectures was further demonstrated by (Krizhevsky et al.,2012) who showed that deep learning models could significantly exceed old methods in large scale image recognition tasks tasks these advancements have encouraged the application of CNN-based models in healthcare where they have been successfully used for detecting disease against chest X-ray images (Raipurkar et al.,2017)

Despite these advancements several challenges remain when adopting DL techniques to medical imaging one of the main challenges in this opportunity & condition of data medical dataset are often limited in size and may contain inconsistencies such as duplicate or corrupted images which can negatively affect model performance in addition selecting an appropriate model architecture is not always straightforward while complex models such as pretrained network are often assumed to perform better this not always the case particularly when working with smaller datasets furthermore there is an increasing need for clarity and interpretation in AI systems used in healthcare as decisions must be explainable and trustworthy (Litjens et al.,2017)

This project aims to investigate the weights of different DL approaches for classified chest X ray images from 3 classes : COVID19 Normal & Pneumonia study begins with data preprocessing including the removal of duplicate and corrupted images to ensure data quality the dataset then splits into training validation & testing sets to allow for reliable decision of model performance

Several models are implemented and compared in this study a baseline CNN model is first developed to establish a reference point this is followed by a tuned CNN model where hyperparameters are adjusted in an attempt to evaluate performance in addition a pretrained DenseNet121 model used to explore the benefits of transfer learning (TL) & further fine tuning is applied assess whether additional optimisation can enhance results TL has been generally approved in medical imaging applications as it allows models to leverage learning from big scale datasets and better allocation on smaller domain specific datasets (Pan and Yang 2010) the achievement of the models checked using metrics such as accuracy loss confusion matrix & classification report in addition Grad CAM is applied to provide visual explanations model indicators recommending the regions of X ray images that commit most of the classification decision (Selvaraju et al., 2017) this elevate model interpretable and addresses a key requirement for the adoption of AI in clinical settings.

Overall this project aims to not only compare different DL models but also to understand how factors such as model complexity data quality and expandability influence performance in medical image classification. The findings are expected to bring beneficial insights into suitability of these approaches for real world healthcare applications.

1.2 Objectives & Research Questions

1.2.1 Aim of Project

The plan is to establish a DL based allocation of chest X ray images into 3 types: COVID19 Normal & Pneumonia. The class uses CNNs & transfer learning techniques to acquire high classification accuracy in addition Grad Cam is adapted to provide visual explanations of model predictions improving interpretability & supporting the transparency of deep learning models in medical applications .

1.2.2 Research Questions : The project is aimed by following research questions

- Does CNN based model exactly allocate chest X ray images to COVID19 Normal & Pneumonia classes ?
- Does the use of transfer learning improve classification performance when compared to a baseline CNN model ?
- Can Grad CAM provide meaningful visual justifications for the predictions built by deep learning models ?
- What are the limitations of CNN based models in detecting COVID19 from chest X ray images ?

1.3 Objectives : To address the research questions the following objectives are defined

- To preprocess the dataset by removing duplicate and corrupted images and preparing it for model training.
- To design & implement a baseline CNN model multi class allocation of chest X ray images.
- Apply hyperparameter tuning techniques to improve achievement of the CNN model.
- To implement a pretrained DenseNet121 model using transfer learning and evaluate its performance.
- To apply fine tuning techniques to the DenseNet121 model and analyse their impact on classification accuracy.
- To classify and analyze the performance of all prototypes using appropriate metrics such as accuracy loss confusion matrix & classification report .
- To implement Grad Cam in order to visualise and interpret model predictions.
- To analyse the strengths & limitations of different deep learning approaches for medical image classification.

1.4 Background and Motivation :

The COVID19 pandemic produced a vital need for fast & scalable diagnostic tools although RT-PCR testing remains standard method for COVID19 detection it can be time consuming & resource intensive specially in high demand situations chest X-ray imaging offers a faster & more accessible solution especially in resource limited settings.

Recent approach in deep learning especially CNNs has shown promising results in medical image resolution including disease detection from radiographic images however many deep learning models operate as Black Box accomplishing it for a clinicians to trust their predictions this has led to increased interest in explainable AI techniques such as Grad-CAM which provide visual insight into model decisions and improve transparency.

Although previous studies have reported high accuracy in COVID19 detection many focus primarily on binary classification tasks (eg.,COVID19 vs Normal) in contrast this project addresses a more realistic multi-class classification problem by distinguishing between COVID19 Normal & Pneumonia cases thereby providing a more comprehensive evaluation of model achievement .

Chapter 2 : Literature Review

2.1 Literature Review

Deep learning has become a powerful advance in medical image reasoning particularly for disease detection using radiographic images among the different techniques CNNs shown strong performance due to capability to learn dimensional features from images (LeCun et al.,2015) CNNs eliminate need for manual feature extraction & been widely applied image allocation task including healthcare applications .

Early work by Krizhevsky et al.(2012) concluded effectiveness of deep CNN big scale image allocation tasks achieving significant improvements over traditional machine learning approaches this success encouraged researchers to apply CNN based models in medical imaging where accurate and automated and highlighted their capability to learn hierarchical features making them suitable for complex image datasets such as chest X rays.

The context of medical imaging Rajpurkar et al.(2017)received ChexNet a deep CNN model trained on chest X ray images to identify pneumonia their model achieved performance comparable to radiologists demonstrating the potential of DL in assisting clinical diagnosis this study provided strong evidence that CNN based models can effectively analyse chest X-ray images and detect abnormalities with high accuracy.

The emergence of the COVID19 pandemic led to increased research into automated detection methods using chest X ray images. Apostolopoulos and Mpesiana (2020) applied transfer learning (TL) using pretrained CNN models to classify COVID19 cases achieving high accuracy even with limited datasets. Their findings suggest that transfer is particularly useful in medical applications where large labelled datasets are not always available.

Similarly Ozturk et al.,(2020) proposed a DL model COVID19 detection using chest X ray images focused on both binary and multi class classification the result demonstrated deep learning models can successfully distinguish between COVID19 Normal & Pneumonia although performance may differ depending on dataset quality & size these studies highlight the importance of evaluating models in multi class settings as real world diagnosis often involves multiple possible conditions.

Transfer learning has been widely adopted to improve model performance and reduce training time model such as DenseNet,ResNet & VGG have been pre trained on big datasets like imagenet & then fine tuned for medical imaging task DenseNet in particular has shown strong performance due to its dense connectivity pattern which improves feature reuse and gradient

flow (Huang et al.,2017)this makes it especially suitable for medical image classification tasks where subtle difference between classes must be captured.

Despite the big accuracy achieved by many DL models a key limitation is their lack of interpretability in clinical settings it is important not only to make accurate predictions but also to understand how those predictions are made this guided to the development of explainable AI (XAI) techniques Grad CAM proposed by selvaraju et al.,(2017) is one of the most extensively used methods for visualising the regions of an image that impact model predictions by highlighting important areas in chest Xray images Grad-CAMhelps improve trust and transparency in deep learning models.

Although previous studies have demonstrated strong performance in COVID19 detection many focus on binary classification problems such as distinguishing COVID19 Pneumonia & Normal cases provides a more realistic and challenging model performance.

In conclusion the literature shows that CNNs and transfer learning models are highly effective for medical image analysis tasks however challenges remain in terms of interpretability dataset quality and generalisation this project builds upon existing research by comparing baseline CNN and TL models while also incorporating Grad-Cam to enhance model interpretability.

Chapter 3 : Methodology

3.1 Dataset Description

The datafile of chest Xray is publicly available consisting of three categories: COVID19 Normal & Pneumonia. This dataset initially contained 5,228 images distributed across the three categories these images represent real world radiographic data used for detecting respiratory conditions

Dataset Link : https://data.mendeley.com/datasets/dvntn9yhd2/1

Table 1: Class Distribution of chest X-ray dataset

CLASS	NUMBER OF IMAGES
COVID - 19	1626
NORMAL	1802
PEUMONIA	1800

3.2 Data Preprocessing :

Before building any models the dataset was carefully explored and cleaned to ensure it was suitable for training

Initially the folder structure was checked to confirm that images were correctly organised into the 3 classes which are COVID-19 Normal & Pneumonia. The number of images in each class was counted which showed a slight imbalance with pneumonia having more samples compared to COVID19.

All images were then checked for corruption by attempting to open each file. No corrupted images were found which meant the dataset was already in a good condition.

One important issue identified during preprocessing was the presence of duplicate images to address this an MD5 hashing approach was used to generate a unique identifier for each image if two images had the same hash value they were treated as duplicate in total 229 duplicate images were detected and removed this step was necessary to avoid data leakage where the same image can be appeared in both training and test sets.

After removing duplicates the dataset size reduced to 4,999 images the updated distribution showed that pneumonia still had the highest number of samples while COVID19 remained slightly underrepresented .

Another observation was that image sizes were not consistent across the dataset for example some images were 224×224 while others were 256×256 so all images are resized to 224×224 pixels to ensure stability & unity with the CNN models.

The cleaned dataset splits into training validation & testing sets using a 70:15:15 ratio.This split is chosen to provide enough data for training while still maintaining divide datasets for validation and testing.

To improve model generalisation data augmentation methods were used such as rotation,zoom & horizontal,flipping was implemented to the training data this helped the model learn more robust features & reduced the risk of overfitting.

Since the dataset showed slight class imbalance class weights were also applied during training this ensured that the model paid more attention to underrepresented classes particularly COVID19 rather than being biased towards the majority class.

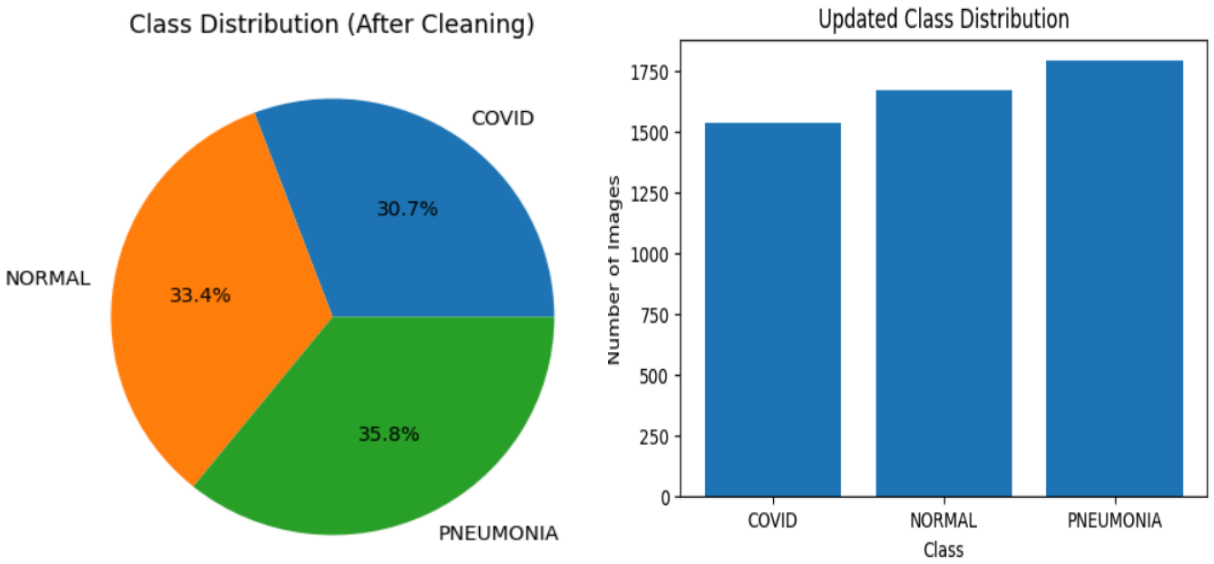


Figure 1:Dataset class Distribution

Chapter 4 : Model Development

After preparing the dataset different deep learning models were developed and tested to understand which approach performs best for this classification problem

4.1 Baseline CNN Model

The first model developed was basic CNN which was used as starting point for comparison

This model abide of 3 convolutional layers with increasing filters (32,64,128) tracked by max pooling layers to less spatial dimension of the feature maps. After convolutional layers the output was flattened & passed through a layer with 128 neurons. A dropout layer (0.5) was added to reduce overfitting. The final O/P layer was the softmax activation function to classify the images into 3 types .

The prototype was shared using Adam optimizer & categorical cross,entropy loss function accuracy was used a main evaluation metrics

The baseline model was useful in understanding how a simple CNN model perform on the dataset before applying more advanced techniques

4.2 Training strategy

The model was trained using training dataset during performance was monitored on the validation set

Two important techniques were used during training :

Early stopping was tested to stop training when validation loss stopped developing which helped to prevent overfitting . Reduce LronPlateau was used to reduce learning rate when the model performance stopped improving allowing better fine tuning of weights

These callback helped stabilise training and developed the overall performance of the model

4.3 Hyperparameter Tuning

To improve the performance of the baseline CNNs hyperparameter tuning was performed using kerastuner with a random search approach

The following parameter were tuned :

Number of filters in convolutional layers number of units in the dense layer dropout rate learning rate

Although tuning was expected to improve performance the results showed that the tuned model did not perform better than the baseline CNN this suggests that increasing model complexity does not always abide to better results especially when baseline model is already performing well.

Table 2 : Hyperparameters Configuration for Tuned Models

Hyperparameter	Tuned CNN Model	Tuned DenseNet121 model (classification head)
Conv_1_filters	32	Not applicable
Conv_2_filters	64	Not applicable
Use _third_conv	False	256
Dense_units	256	0.2
Dropout_rate	0.2	0.2
Learning_rate	0.001	0.01

4.4 Transfer Learning Using DenseNet121

To further improve performance a transfer learning approach was used by implementing the DenseNet121 model pre-trained on image Net the original classification layer was replaced with a custom output layer suitable for the three class classification task the earlier layers of the network were initially frozen so that prelearned features could be retained

Using TL significantly reduced training time and allowed model to learn more complex features compared to the baseline CNN

4.5 Fine-tuning

After initial training fine tuning was applied to the DenseNet121 model by unfreezing some of the deeper layers & retraining them with a lower learning rate.

This allowed model to accommodate better to specific features of chest X ray images during this process same callbacks (early stopping & reducelronplateau) were used to control overfitting & optimise learning

Fine tuning resulted in improved performance compared to the standard transfer learning model showing that adjusting pre trained weights can provide additional benefits

4.6 Use Of Class Weights

Since the dataset was slightly imbalanced class weights were applied during model training ensured that model did not become biased towards majority class (pneumonia) & gave appropriate importance to minority classes such as COVID19 this step was particularly useful in improving recall for underrepresented classes

Chapter 5 : Results and Discussion

After training and evaluating all models their performance was assessed using the test dataset the results show clear difference between the approaches used.

The overall test accuracy & loss for each model are summarised as follow :

5.1 Baseline CNN Performance

The baseline CNN achieved highest test accuracy of 97.47% with strong performance despite its relatively simple architecture. This suggests that the dataset was well performed & contained distinguishable features across the three classes .

The confusion matrix & classification report further confirmed that the model performed consistently across COVID-19 Normal & Pneumonia classes with high precision & recall values. The low test loss also indicates stable model predictions.

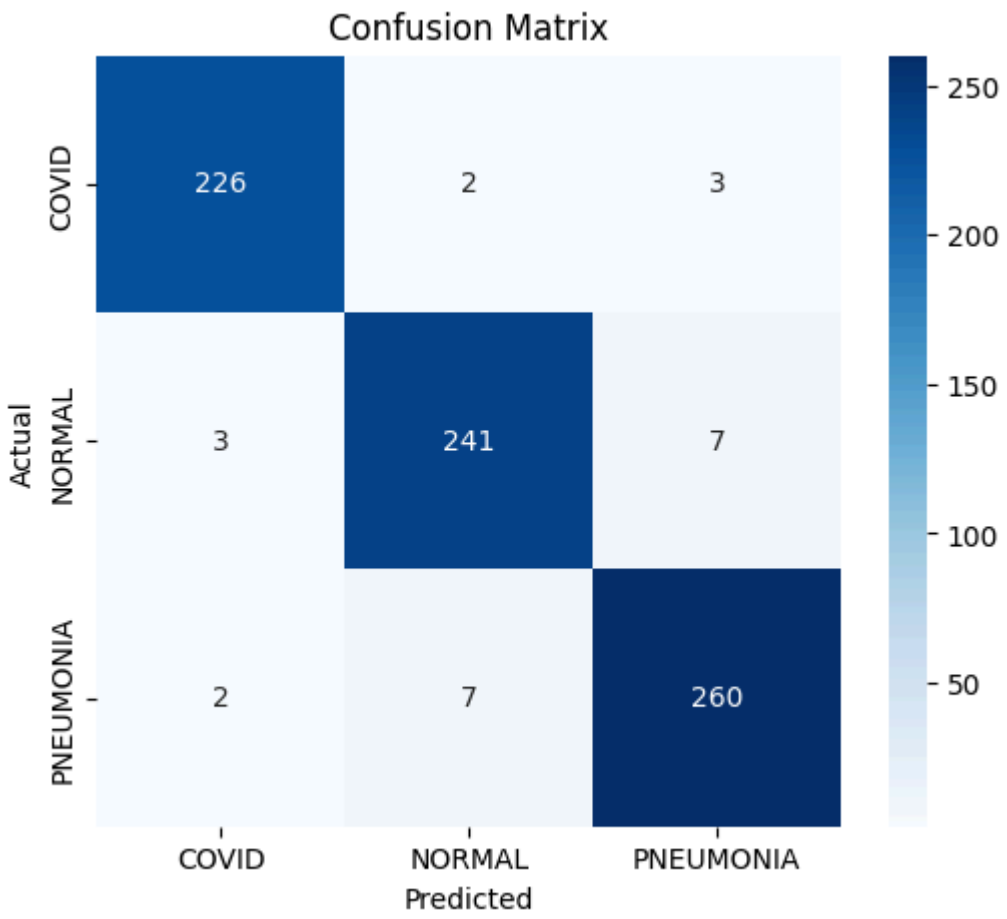


Figure 1: Confusion Matrix for Baseline CNN Model

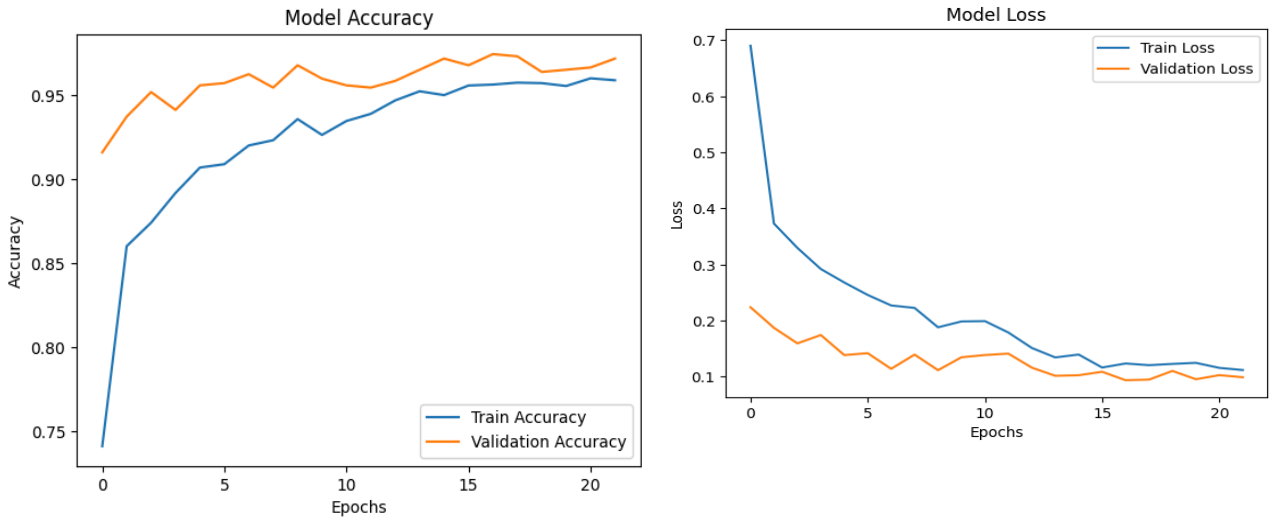


Figure 1.1 : Base line CNN Model Accuracy Model Loss

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_5 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_5 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_1 (Flatten)	(None, 86528)	0
dense_2 (Dense)	(None, 128)	11,075,712
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 3)	387

Total params: 11,169,347 (42.61 MB)

Trainable params: 11,169,347 (42.61 MB)

Non-trainable params: 0 (0.00 B)

5.2 Tuned CNN Performance

In contrast the tuned CNN achieved a lower accuracy of 89.08% which was significantly below the baseline model

Although hyperparameter tuning is generally expected to improve performance the results suggest that the selected parameter combinations may not have been optimal it is also possible that increasing model complexity led to reduced generalisation this highlights that hyperparameters tuning must be carefully controlled as it does not always guarantee improved results

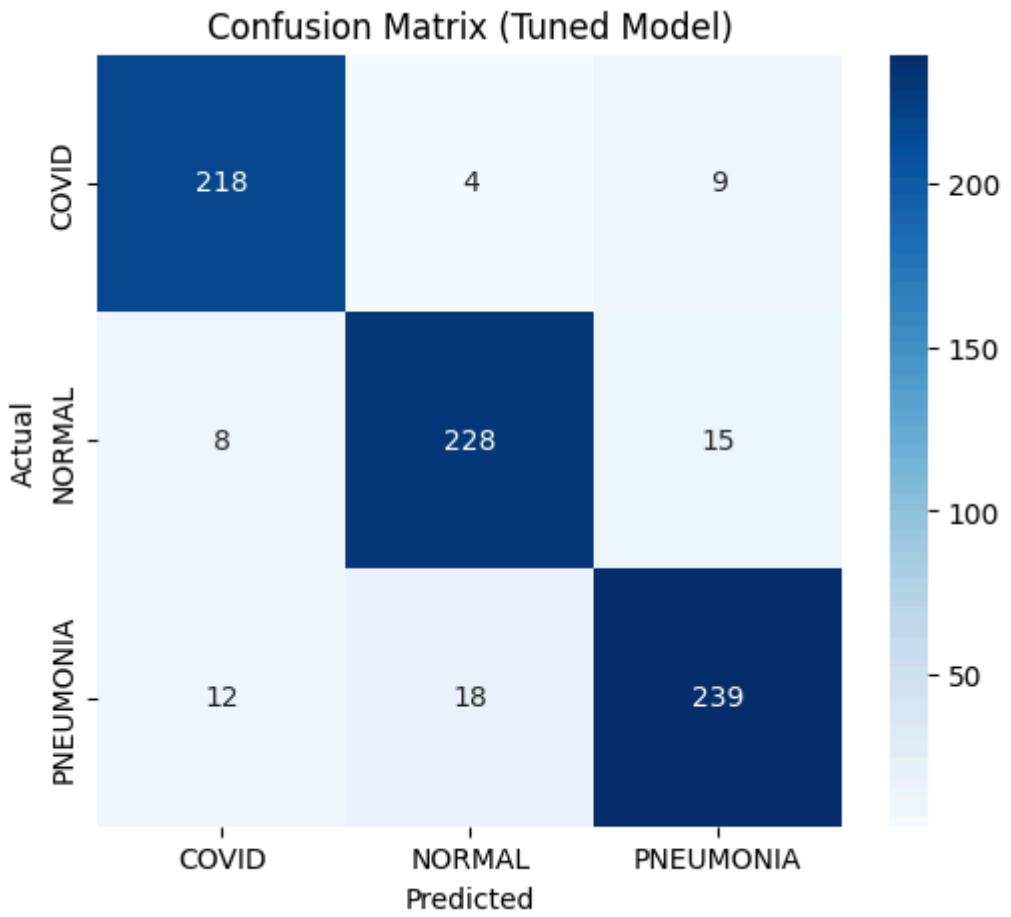


Figure 2 : Confusion Matrix for tuned CNN Model

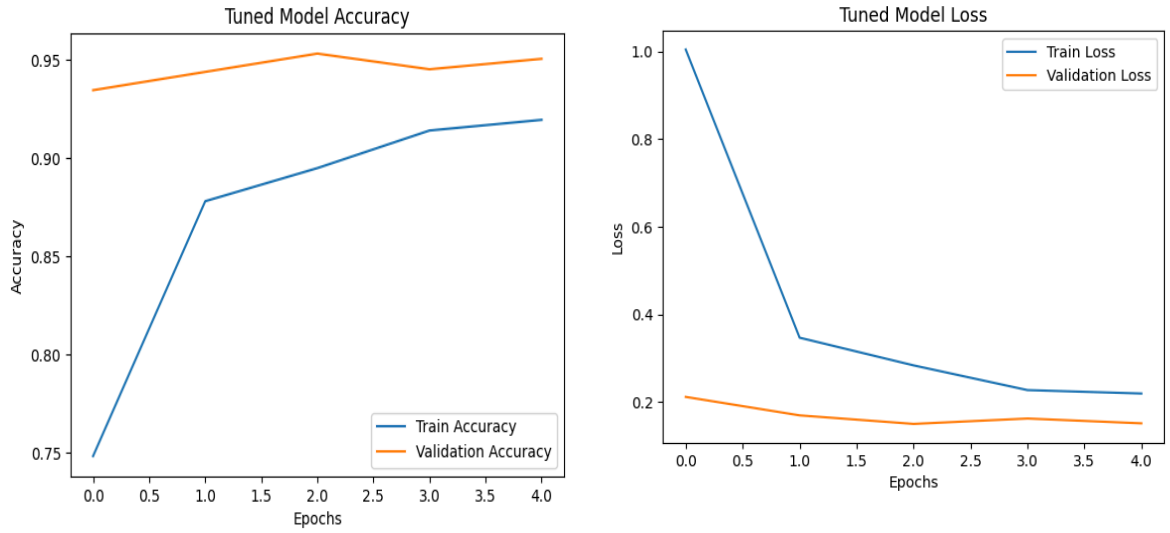


Figure 2.1 : Tuned CNN Model Accuracy Model Loss

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0
flatten_1 (Flatten)	(None, 186624)	0
dense_2 (Dense)	(None, 256)	47,776,000
dropout_1 (Dropout)	(None, 256)	0
dense_3 (Dense)	(None, 3)	771

Total params: 47,796,163 (182.33 MB)

Trainable params: 47,796,163 (182.33 MB)

Non-trainable params: 0 (0.00 B)

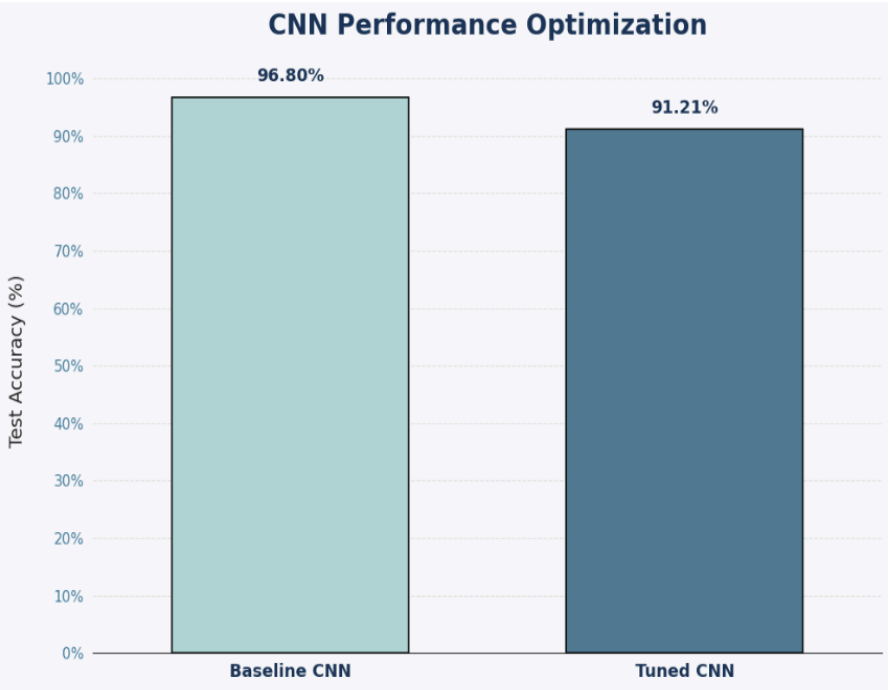


Figure 2.2:Comparison Bar Plot For Baseline CNN Tuned CNN

5.3 DenseNet121 Performance

The DenseNet121 model using TL achieved an accuracy of 96.01% demonstrating strong performance TL approved the pattern to leverage pre trained features which is particularly useful when working with limited datasets however the model did not outperform the baseline CNN suggesting that the task may not require highly complex architectures

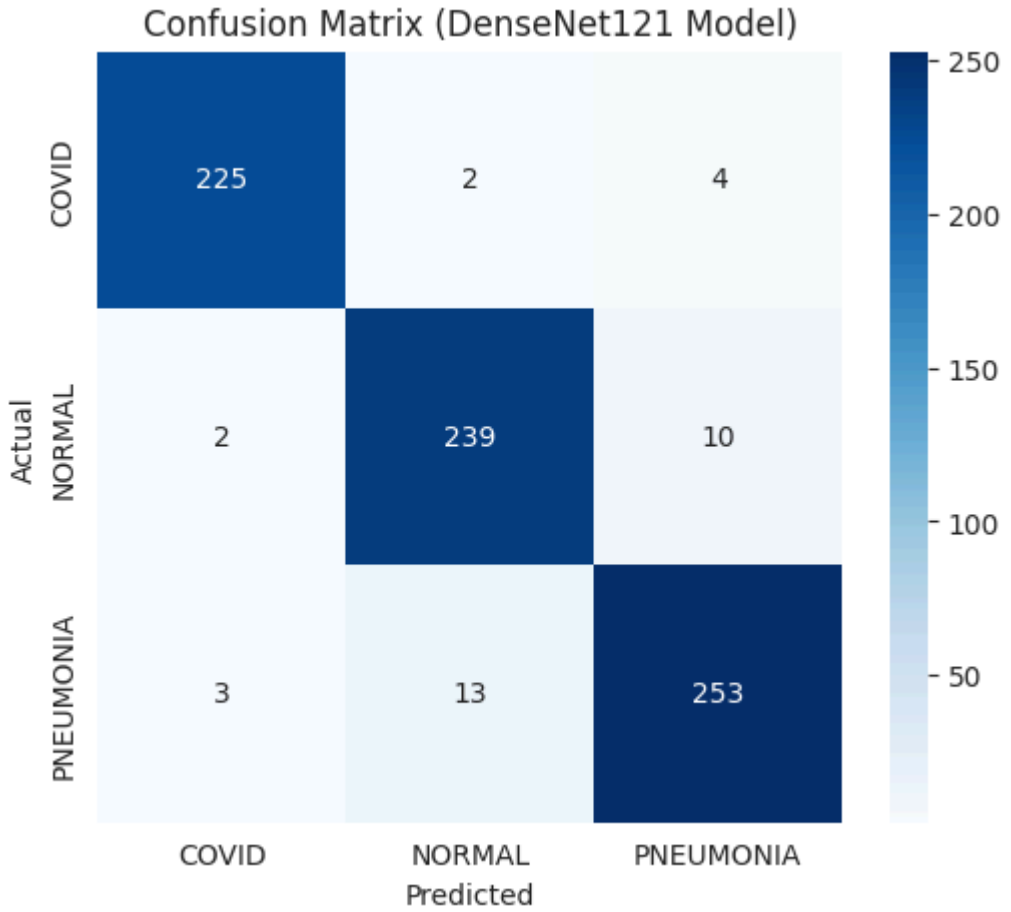


Figure 3 : Confusion Matrix for Densenet121 Model

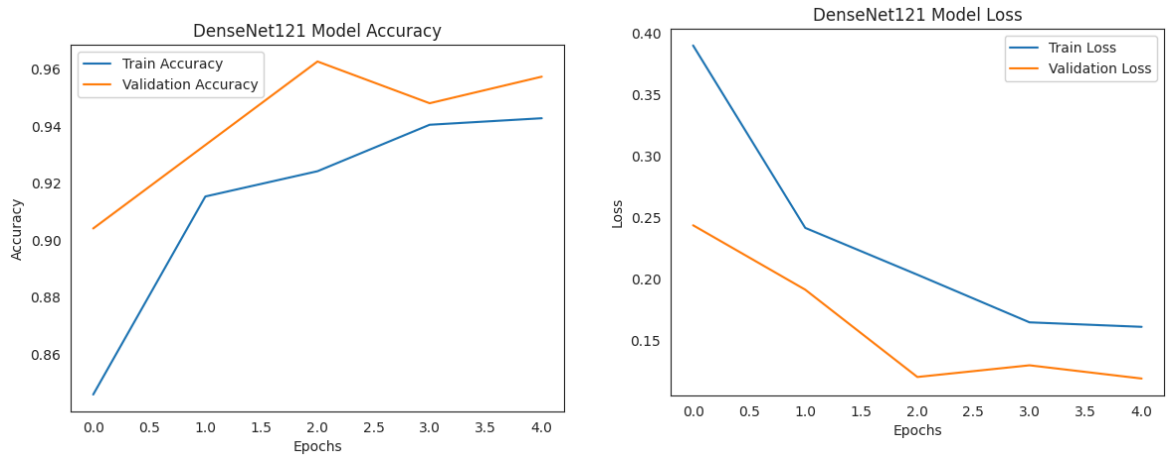


Figure 3.1 : DenseNet121 Model Accuracy Model Loss

Model: "sequential_3"

Layer (type)	Output Shape	Param #
densenet121 (Functional)	(None, 7, 7, 1024)	7,037,504
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1024)	0
dense_6 (Dense)	(None, 256)	262,400
dropout_3 (Dropout)	(None, 256)	0
dense_7 (Dense)	(None, 3)	771

Total params: 7,300,675 (27.85 MB)

Trainable params: 263,171 (1.00 MB)

Non-trainable params: 7,037,504 (26.85 MB)

5.4 Tuned DenseNet121 Performance

After applying tuned the DenseNet121 model achieved an improved accuracy of 96.54% this improvement indicates allowing selected layers to retrain helped the model adapt better to domain specific looks in chest X ray images although improvement was moderate it confirms the strength of fine tuning in transfer learning

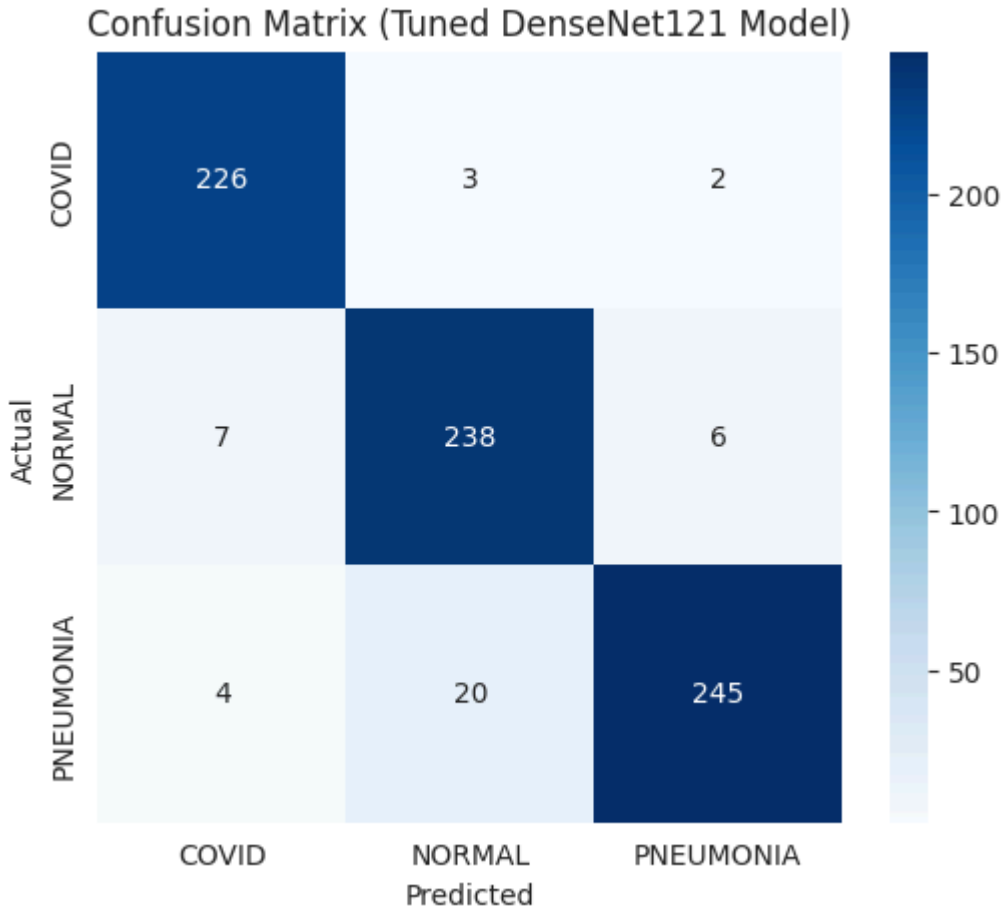


Figure 4 : Confusion Matrix for Tuned DenseNet121 Model

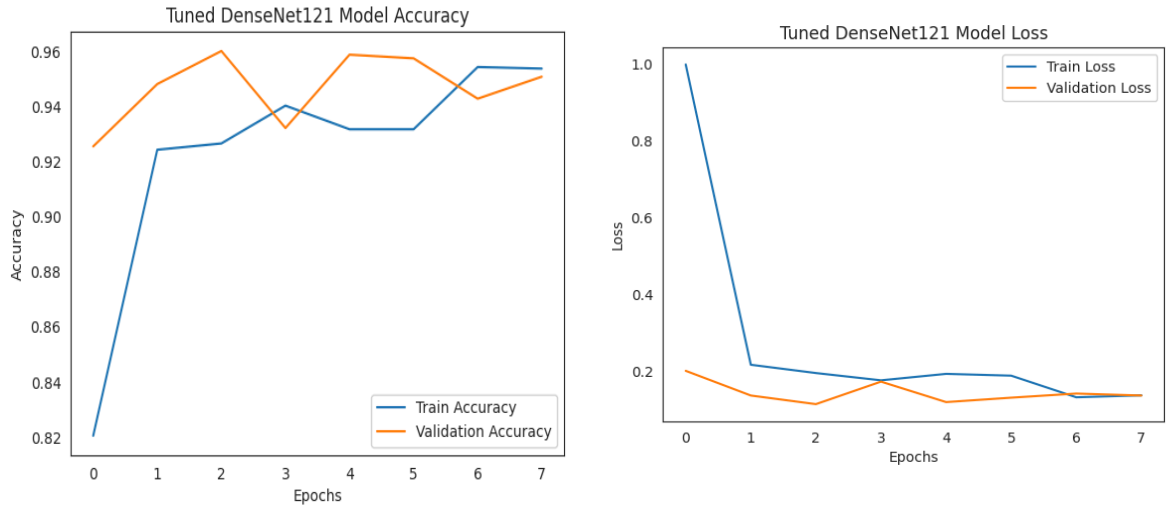


Figure 4.1 : Tuned DenseNet121 Model Accuracy Model Loss

Model: "sequential_1"

Layer (type)	Output Shape	Param #
densenet121 (Functional)	(None, 7, 7, 1024)	7,037,504
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1024)	0
dense_2 (Dense)	(None, 256)	262,400
dropout_1 (Dropout)	(None, 256)	0
dense_3 (Dense)	(None, 3)	771

Total params: 7,300,675 (27.85 MB)

Trainable params: 263,171 (1.00 MB)

Non-trainable params: 7,037,504 (26.85 MB)

Table 3 : Comparative performance models on test set

Model name	Test loss	Test accuracy
Baseline CNN	0.132	0.968
Tuned CNN	0.244	0.912
DenseNet121 (untuned)	0.145	0.955
DenseNet121 (tuned)	0.165	0.944

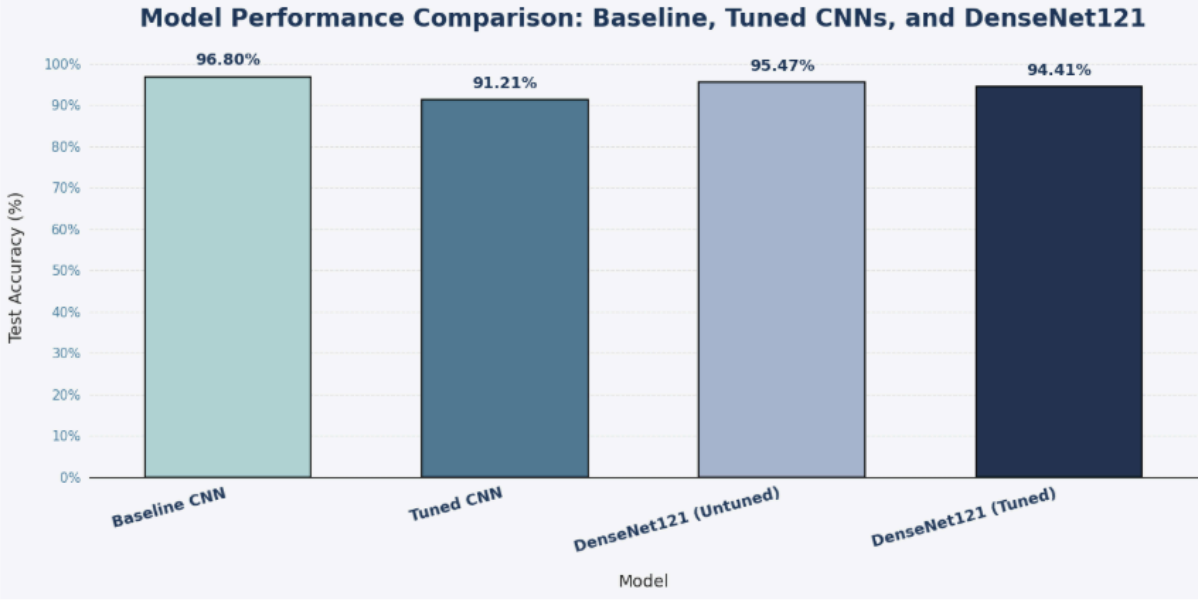


Figure5: `Model Performance Comparison: Baseline, Tuned CNNs, and DenseNet121`

5.5 Training Behaviour

Analysis of training and validation curves showed stable convergence across most models the use of early stopping prevented overfitting training when validation loss no longer improved while reduced lronplateau allowed better optimisation during later training stages overall there was no significant evidence of overfitting indicating that data preprocessing and augmentation were effective

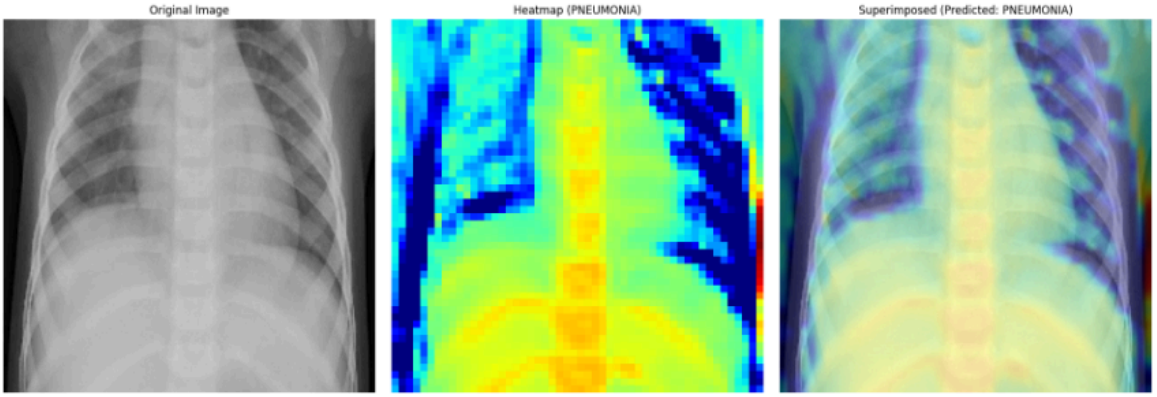
5.6 Explainability Using Grad CAM

Grad-CAM adapted to anticipate the regions of images that changed model predictions the resulting heatmaps showed that the model focused primarily on lung regions which are clinically relevant areas for identifying respiratory diseases this suggest that the model learned meaningful features rather than relying on irrelevant patterns

```
Upload an image (JPG, PNG, JPEG) to get a prediction and Grad-CAM visualization for the best pe
rforming model (Baseline CNN):
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving 12341234.png to 12341234.png
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step
Predicted: PNEUMONIA (Confidence: 99.96%)
Using last convolutional layer: conv2d_5
```



5.7 Discussion

This results indicate that model performance is strongly influenced by data quality and preprocessing the baseline CNN achieved the highest accuracy demonstrating that simpler models can perform effectively when the dataset is well prepared

In contrast the tuned CNN performed worse highlighting that increasing complexity does not necessarily lead to better outcomes the transfer learning models performed well with fine tuning providing additional improvement although not exceeding the baseline model these findings are consistent with existing literature where both CNNs and transfer learning approaches have shown strong achievement in medical image classification tasks.

Chapter 6 : Conclusion &Future work

6.1 Conclusion

The project planned to develop & classify (DL) models for allocation of chest X ray images into COVID19 Normal & Pneumonia categories. Multiple approaches were explored including a baseline CNN a tuned CNN and transfer learning using DenseNet121 with and without fine tuning .

The results demonstrate that all models were capable of achieving high classification with test accuracies above 89% amid the models the baseline CNN got the highest accuracy of 97.47% outperforming both the tuned CNN & the TL models indicates that for this dataset a relatively simple CNN architecture was sufficient to capture the distinguishing feature of chest X-ray images.

The tuned CNN did not improve performance highlighting that hyperparameters tuning does not always abide to better results particularly when the baseline model is already well optimised the DenseNet121 model achieved strong performance and fine tuning further improved its accuracy demonstrating benefits of TL & model adaptation.

The use of class weights contributed to balanced model performance across all classes ensuring that minority classes such as COVID19 were not underrepresented during training additionally the application of Grad CAM provided visual insights into model predictions showing that the model intent on clinically related region of lungs this supports the reliability and interpretability of the model .

Overall the findings suggest that data quality and preprocessing play a critical role in model achievement and that simpler models can be highly effective when the dataset is well prepared

6.2 Justification Of Approach

The methodology used in this project was appropriate for the research problem

The use of CNN based model is well supported in literature for image classification tasks particularly in medical imaging transfer learning was selected to leverage pre trained knowledge and improve performance with a moderate dataset size fine tuning was applied to further adapt the model to domain specific features.

The decision to remove duplicate images and apply data augmentation ensured that models are trained on a clean and diverse dataset. The use of class weights addressed class imbalance improving fairness in model predictions.

Evaluation using multiple metrics including accuracy loss confusion matrix and classification report provided a comprehensive assessment of model conduct. The involvement of Grad CAM addressed limitation of model interpretability which is an important consideration in healthcare application.

6.3 Limitations

Despite achieving strong results several limitations were reasoned

- The dataset size was relatively limited compared to big scale medical datasets
- The data may not fully represent real-world variability across different populations
- Only chest X-ray images were used whereas clinical diagnosis often involves multiple data source
- The study focused on three classes which may not cover all possible lung conditions

6.4 Future Work

Based on the findings and limitations several improvements can be suggested for future work :

- Use larger and more diverse dataset to improve model generalisation
- Include additional classes to better reflect real world diagnostic scenarios
- Experiment with other deep learning architectures such as ResNet or EfficientNet
- Apply advanced hyperparameters optimisation techniques for better tuning
- Combine image data with clinical data (eg: patient history,symptoms) for more accurate predictions
- Explore more explainability methods beyond Grad-CAM for deeper model understanding

6.5 Final Remarks

The project demonstrates that DL models such as CNN based approaches can be effectively applied to medical image classification tasks while advanced models such as DenseNet121 offer strong performance. The results highlight that simpler models can achieve comparable or even superior results when supported by proper data preprocessing and training strategies .

The inclusion of explainability techniques further strengthens the practical applicability of such models building more suitable for real world healthcare environments

References

- LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P., 1998. Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), pp.2278–2324.

- Krizhevsky, A., Sutskever, I. and Hinton, G.E., 2012. ImageNet classification with deep convolutional neural networks. Advances in Neural Information Processing Systems, 25.
- Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K. and Lungren, M., 2017. CheXNet: Radiologist-level pneumonia detection on chest X-rays. arXiv preprint arXiv:1711.05225.
- Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., van der Laak, J.A.W.M., van Ginneken, B. and Sánchez, C.I., 2017. A survey on deep learning in medical image analysis. Medical Image Analysis, 42, pp.60–88.
- Pan, S.J. and Yang, Q., 2010. A survey on transfer learning. IEEE Transactions on Knowledge and Data Engineering, 22(10), pp.1345–1359.
- Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D., 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localisation. Proceedings of the IEEE International Conference on Computer Vision, pp.618–626.
- Apostolopoulos, I.D. and Mpesiana, T.A. (2020) 'Covid-19: automatic detection from X-ray images utilizing transfer learning with convolutional neural networks', Physical and Engineering Sciences in Medicine, 43(2), pp. 635–640.
- Huang, G., Liu, Z., Van Der Maaten, L. and Weinberger, K.Q. (2017) 'Densely connected convolutional networks', Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 4700–4708.
- Krizhevsky, A., Sutskever, I. and Hinton, G.E. (2012) 'ImageNet classification with deep convolutional neural networks', Advances in Neural Information Processing Systems, 25, pp. 1097–1105.
- LeCun, Y., Bengio, Y. and Hinton, G. (2015) 'Deep learning', Nature, 521(7553), pp. 436–444
- Ozturk, T. et al. (2020) 'Automated detection of COVID-19 cases using deep neural networks with X-ray images', Computers in Biology and Medicine, 121, p. 103792.
- Rajpurkar, P. et al. (2017) 'CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning', arXiv preprint arXiv:1711.05225.
- Rawat, W. and Wang, Z. (2017) 'Deep convolutional neural networks for image classification: A comprehensive review', Neural Computation, 29(9), pp. 2352–2449.
- Selvaraju, R.R. et al. (2017) 'Grad-CAM: Visual explanations from deep networks via gradient-based localisation', Proceedings of the IEEE International Conference on Computer Vision, pp. 618–626.

- Detection of COVID-19 Using Transfer Learning and Grad-CAM Visualization on Indigenously Collected X-ray DatasetMuhammad UmairMuhammad Shahbaz KhanFawad AhmedFatmah BaothmanFehaid AlqahtaniMuhammad Alian Jawad Ahmad https://pmc.ncbi.nlm.nih.gov/articles/PMC8434081/
- COVID-19 Detection Using Deep Learning Algorithm on Chest X-ray ImagesShamima Akter F M Javed Mehedi ShamratSovon Chakraborty , Asif Karim Sami Azam https://pmc.ncbi.nlm.nih.gov/articles/PMC8614951/
- Kumar, Sachin (2022), Covid19-Pneumonia-Normal Chest X-Ray Images Mendeley Data, V1, doi: 10.17632/dvntn9yhd2.1https://data.mendeley.com/datasets/dvntn9yhd2/1